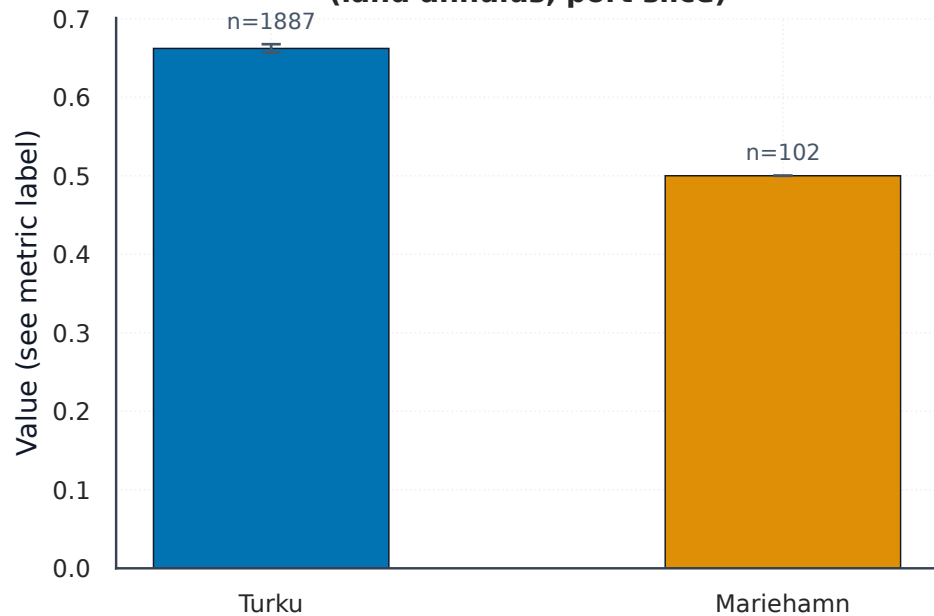
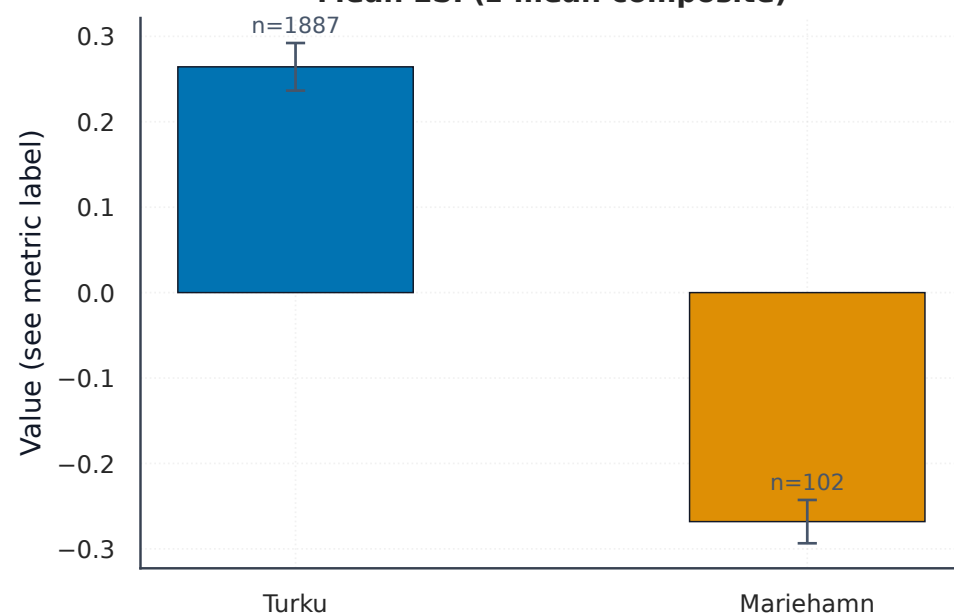


Cross-port urban/coastal structure (shared annulus mask)

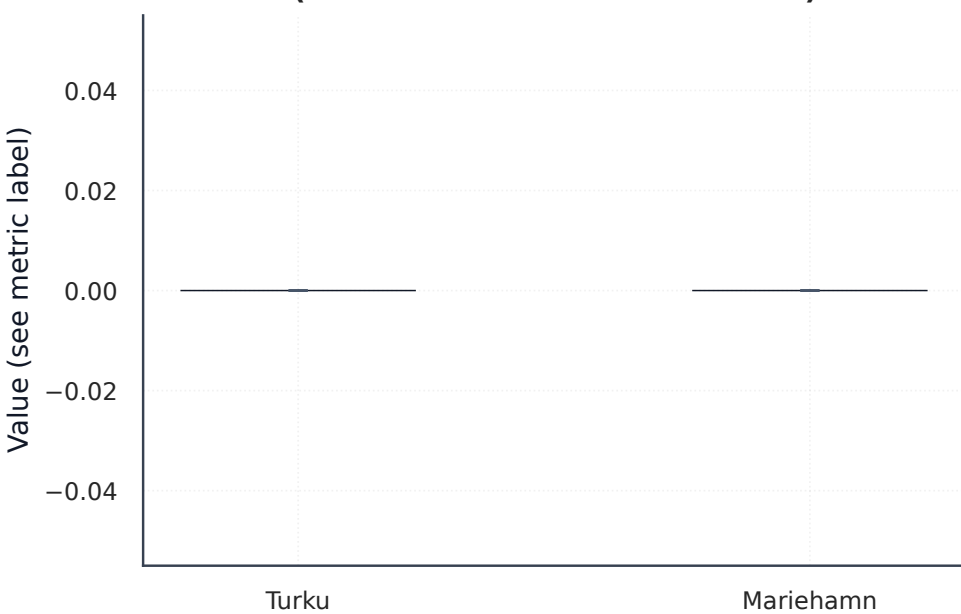
Mean coastal exposure score (land annulus, port slice)



Mean ESI (z-mean composite)



NO₂ persistence (mean 30-50 km ÷ mean 0-3 km)



MEI landward slope (weighted OLS vs band centre, km⁻¹)

